



IILM UNIVERSITY
Greater Noida, Uttar Pradesh

SMARTBRANCH 360

Internship Topic: Designing, Securing & Validating a Branch-Office Network

Cisco Networking Academy — Network Technician Career Path

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PROGRAM & SEMESTER

MCA, 3rd Semester (2025-27)

ORGANIZATION NAME

IILM University – Cisco Virtual Internship Program

What This Presentation Covers

01

Internship Program Overview

Three Cisco Networking Academy modules completed

02

Problem Statement & Objectives

The SmartBranch 360 branch-office design brief

03

Network Design & Build

Topology, VLANs, IP addressing, in Cisco Packet Tracer

04

Security & Verification

Guest isolation, management SSH, live traffic tests

05

Troubleshooting & Tooling

Fault injection framework and a Python assurance tool

06

Key Learnings

What the project taught about real network verification

Program Overview

Cisco Networking Academy's Network Technician Career Path, completed through IILM University

M1

Networking Essentials

OSI/TCP-IP models, IP addressing & subnetting, switching and routing fundamentals.

M2

Getting Started with Cisco Packet Tracer

Building topologies, cabling conventions, real-time vs. simulation mode.

M3

Exploring Networking with Cisco Packet Tracer

VLAN trunking, inter-VLAN routing, wireless APs, and access control lists.



Project Objectives

1 Design a segmented branch-office network

2 Build the topology in Packet Tracer

3 Implement VLAN segmentation and routing

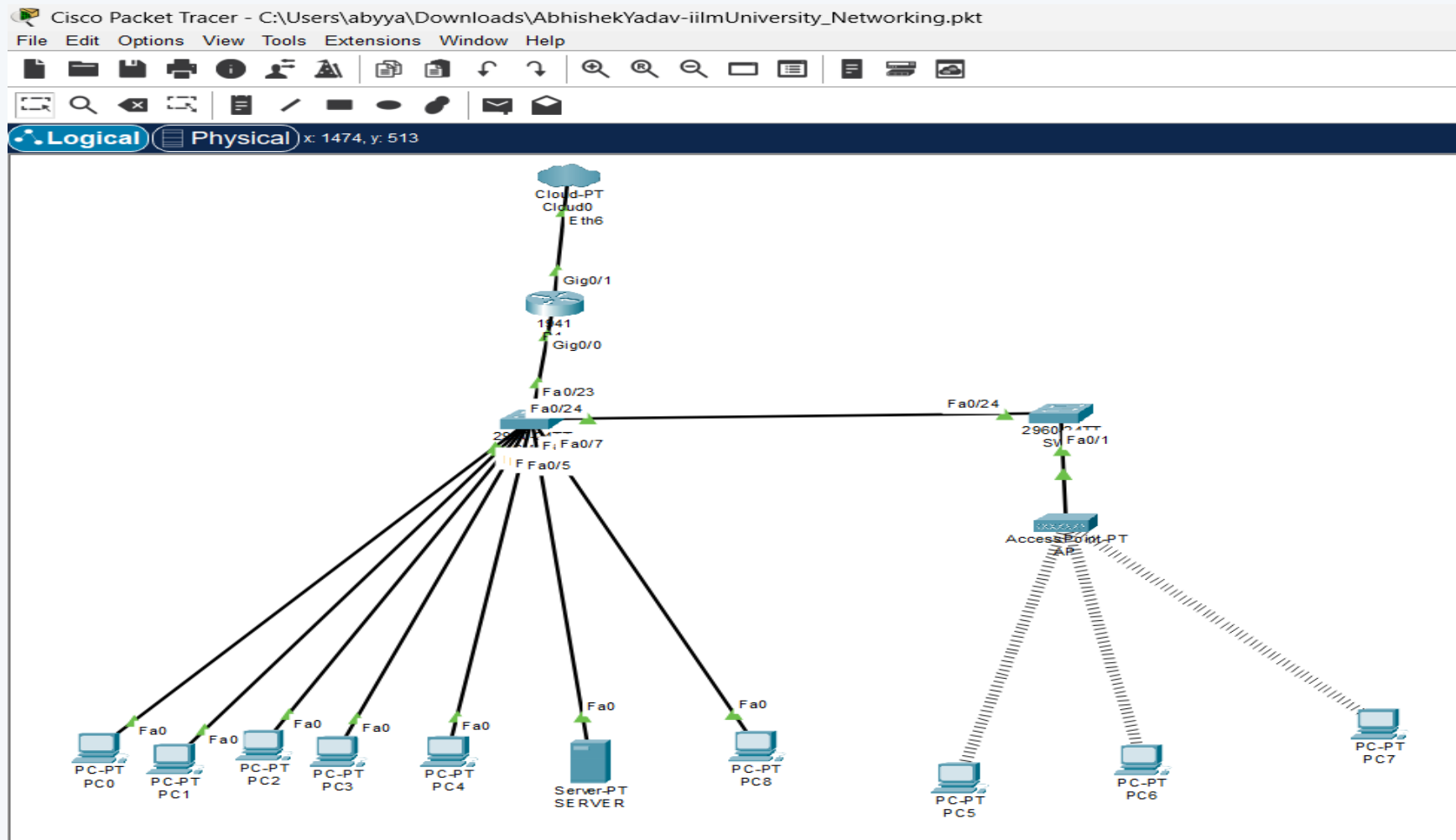
4 Provide automatic addressing (DHCP/DNS)

5 Enforce network security (ACLs, SSH)

6 Practice structured fault troubleshooting

7 Automate configuration checking in Python

Network Topology — Built in Cisco Packet Tracer



1

Router

2

Switches

1

Wireless AP

9

Endpoints

VLAN & IP Addressing Plan

Four logical segments keep guest, employee, server, and management traffic isolated

VLAN	Name	Subnet	Purpose
10	Employee	10.10.10.0/24	Wired employee workstations
20	Guest	10.10.20.0/24	Guest Wi-Fi via wireless AP
30	Server	10.10.30.0/24	Internal server
99	Management	10.10.99.0/24	Device management (SSH access)

Design rationale: Router-on-a-stick sub-interfaces (G0/0.10, .20, .30, .99) route between VLANs, so guest traffic, business-critical server traffic, and administrative access never mix on the wire — the single biggest low-cost security improvement for a small branch office.

Security Configuration



Guest Isolation

Access control list denies Guest VLAN (20) traffic to the Server VLAN (30), while still permitting internet-bound traffic.



Management-only SSH

SSH access to routers/switches is permitted only from the Management VLAN (99); every other SSH attempt is denied.



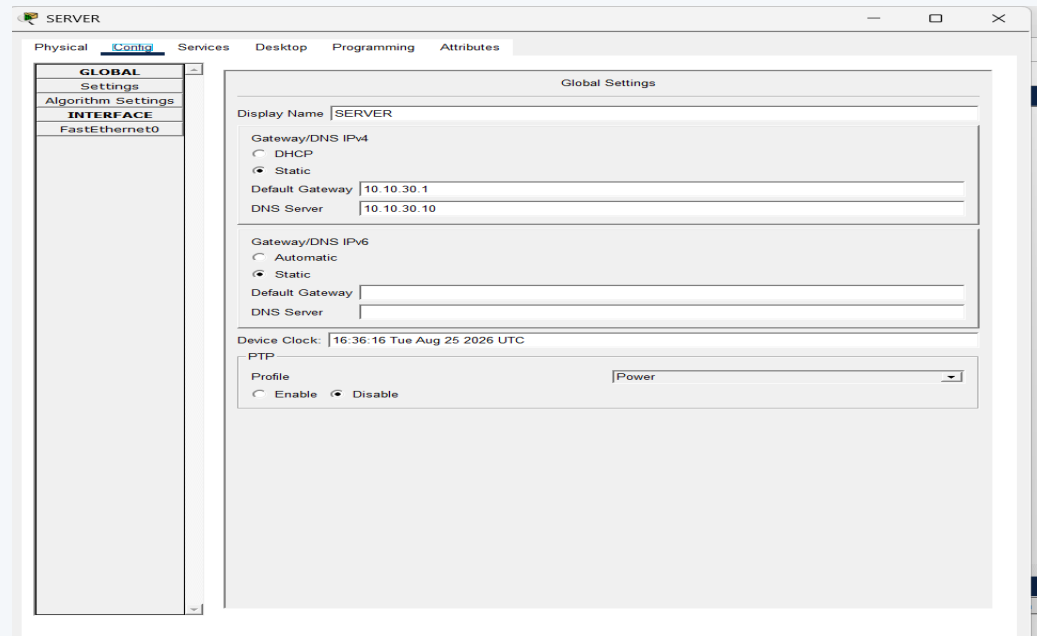
Inter-VLAN Control

Only traffic explicitly required by the design is permitted — e.g., Employee → Server, Employee/Guest → Internet.

Verified on Real Devices — Not Just the Diagram

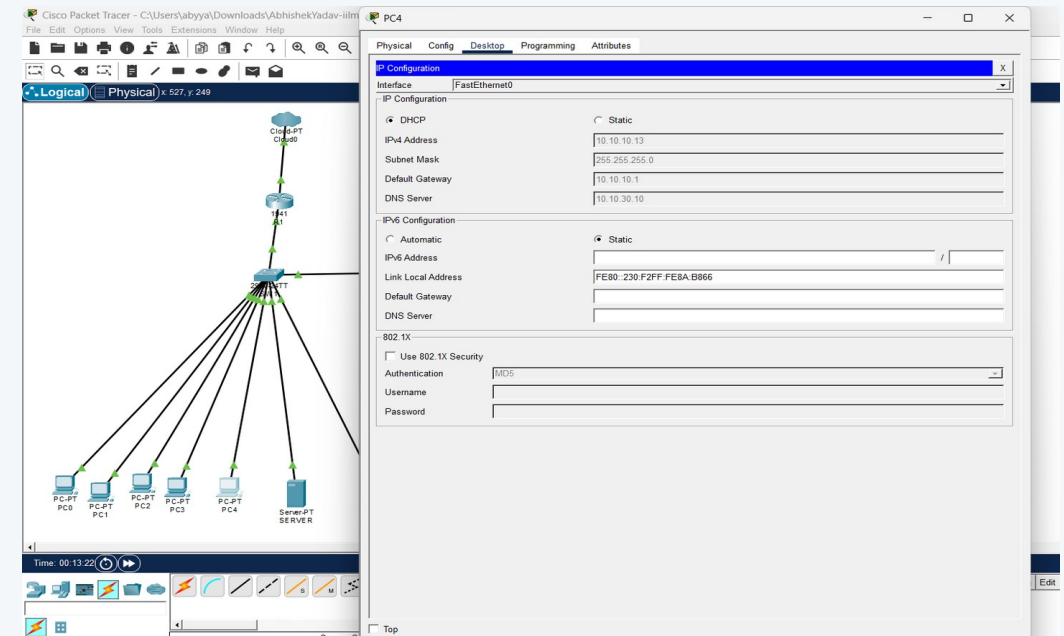
Each device's own IP Configuration was opened and checked against the VLAN/IP plan

Server — Static Configuration



Gateway 10.10.30.1 · DNS 10.10.30.10 — matches Server VLAN 30 exactly

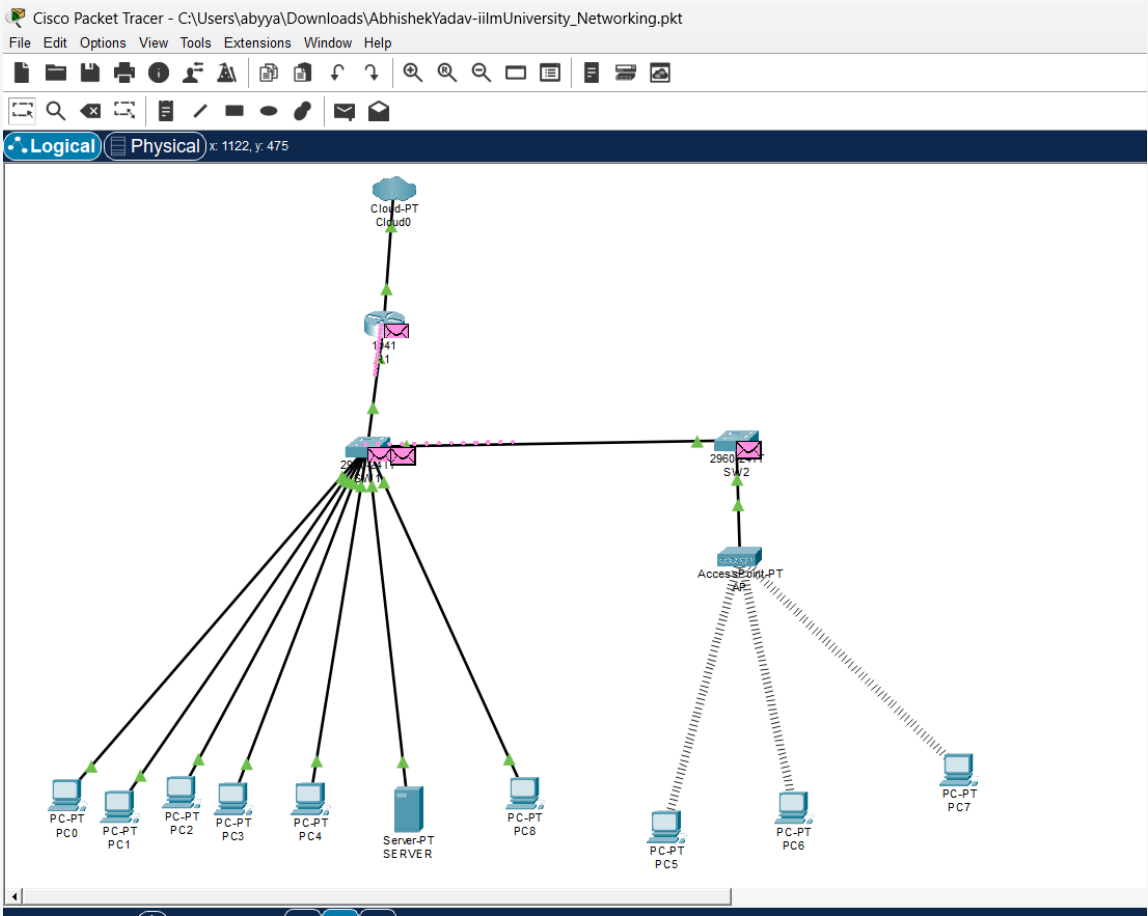
PC4 (Employee VLAN) — DHCP



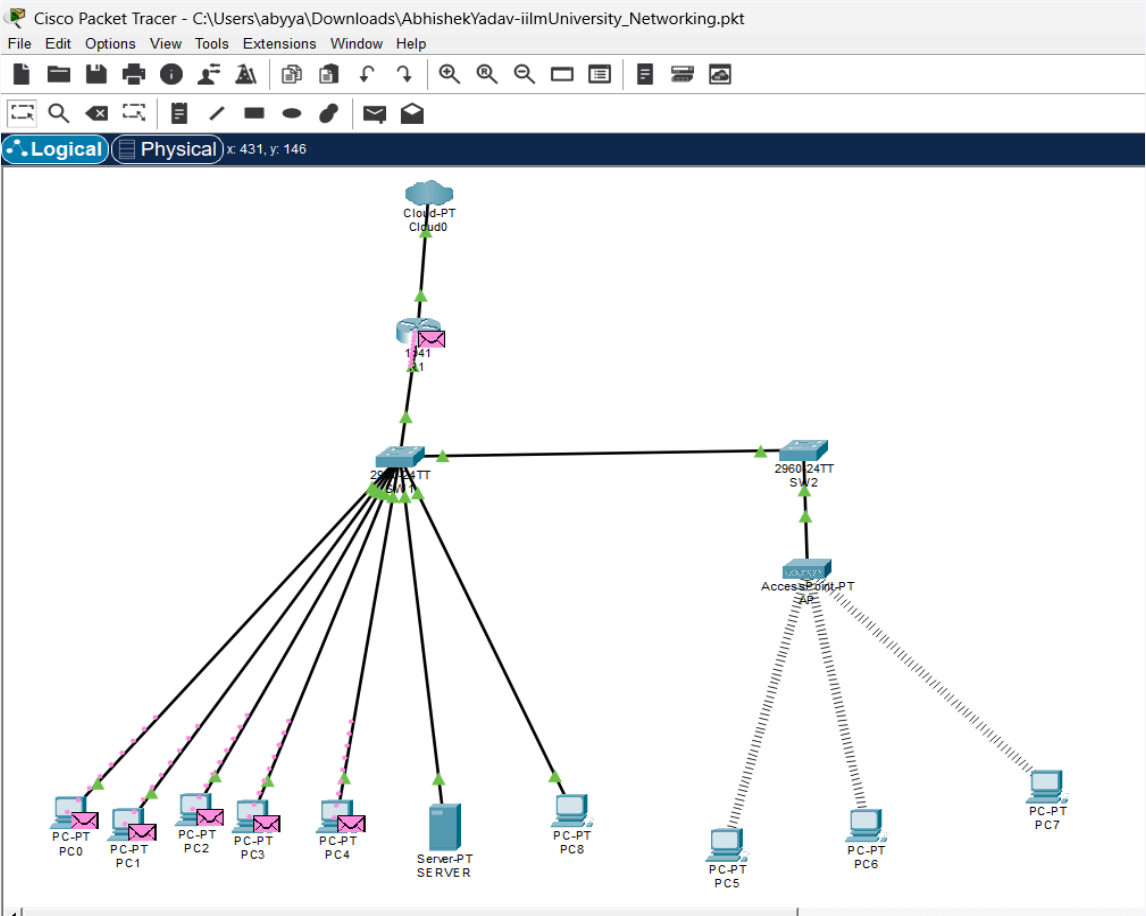
Leased 10.10.10.13, gateway 10.10.10.1, DNS 10.10.30.10 — proves inter-VLAN routing works

Live Traffic Verification (Simulation Mode)

A test PDU was sent through the topology to confirm packets actually traverse the trunk links — not just appear connected on the diagram



PDU in transit: R1 → SW1 → SW2 trunk



PDU reaching Employee VLAN endpoints (PC0-PC4)

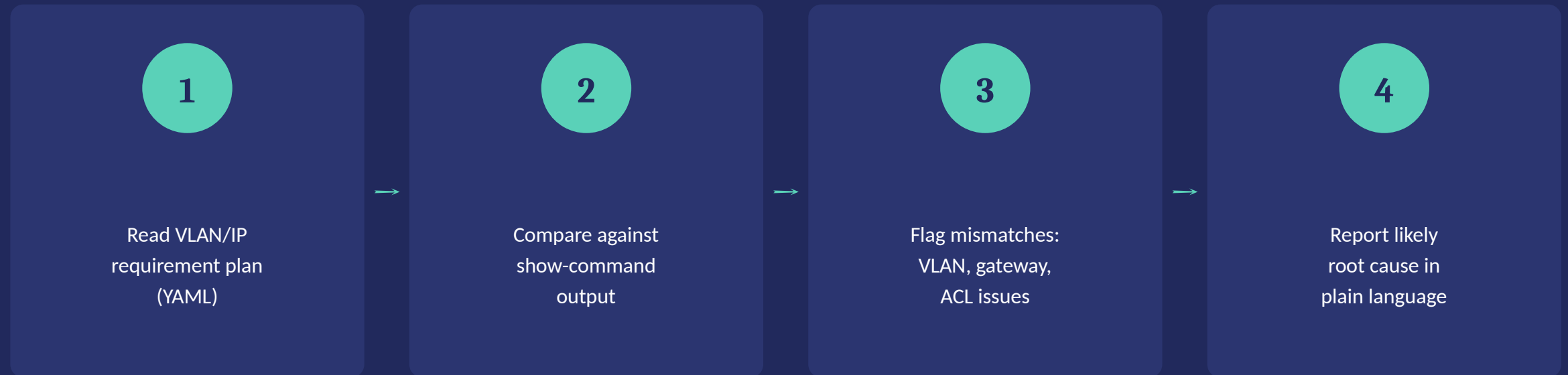
Fault Injection & Troubleshooting

Five faults deliberately introduced, one at a time — practicing symptom → root cause → fix reasoning

#	Fault	Symptom	Fix
1	Missing VLAN on trunk	Guest PC has no connectivity	Add VLAN 20 to trunk's allowed list
2	Wrong default gateway	PC cannot reach other subnets	Correct gateway to match VLAN sub-interface
3	DHCP pool misconfigured	PC fails to obtain an IP	Correct network/mask in DHCP pool
4	ACL blocking DNS	Ping works, name resolution fails	Add a permit rule for DNS in the ACL
5	NAT misconfiguration	Internal hosts can't reach internet	Correctly mark NAT inside/outside

Python Assurance Tool

A lightweight Python script reads the VLAN/IP requirement plan and, optionally, device show-command output — then reports likely configuration problems in plain language, speeding up verification after every change without replacing hands-on testing.



Key Learnings

Verify, don't assume

Open each device's own IP Configuration and read back what it actually believes its address, gateway, and DNS are — instead of trusting the topology diagram.

Static vs. dynamic addressing

Infrastructure devices like servers need static IPs because other devices reference them; workstations are safely left on DHCP.

Segmentation is the cheapest security win

Separating guest, employee, server, and management traffic into VLANs blocks entire classes of problems at near-zero cost.

A network is only as trustworthy as its verification process

Building it right the first time matters less than being able to prove — quickly and repeatably — that it still works after every change.

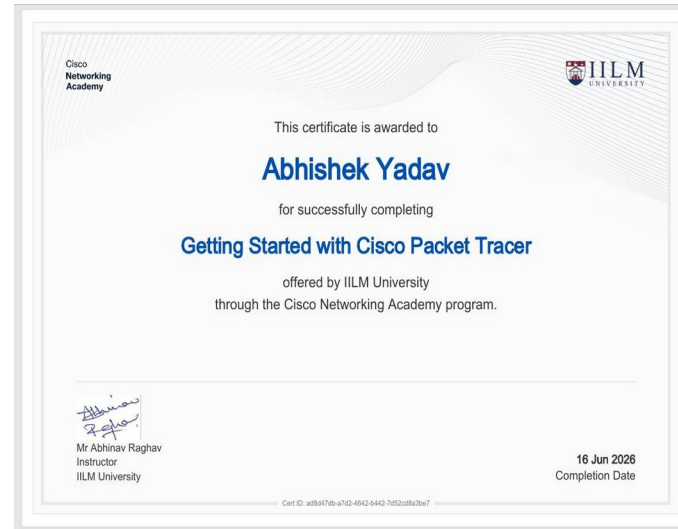
Certifications Earned



Networking Essentials

Cisco Networking Academy · IILM University

Completed: 20 Jun 2026



Getting Started with Cisco Packet Tracer

Cisco Networking Academy · IILM University

Completed: 16 Jun 2026



Exploring Networking with Cisco Packet Tracer

Cisco Networking Academy · IILM University

Completed: 13 Jun 2026

Thank You

SmartBranch 360 demonstrates how VLAN segmentation, routing, DHCP/DNS/NAT, and access control combine into a secure, verifiable branch-office design.

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